



Team Details

- a. **Team name:** ASM FinTech Developer
- b. **Team leader name:** Abhinav Mukwane
- c. **Problem Statement:** Financial Health Score
- d. **Round:** Top-5 Finalist — Final Submission (September 2026)

Brief about the idea

India has 6.4 crore MSMEs contributing ~30% of GDP, yet 80%+ have no formal credit. New-to-Credit (NTC) and New-to-Bank (NTB) enterprises are rejected for missing paperwork — not for being bad borrowers.

FinRiskLensAI — AI-powered MSME Financial Health Score platform (Top-5 Finalist build)

- One consent, five sources: Udyam, GST (7 return types), ITR, bank statements via Account Aggregator, and MCA company & director records
- One score, six pillars: Financial Health Score 0–1000 from a rules engine + ML.NET LightGBM — every score comes with plain-language reasons
- Financial Health Card: score gauge, 6-pillar radar, strengths & risks, lending eligibility (working capital + term loan) and the right product (CGTMSE / MUDRA / standard)
- **NEW — Score History:** every analysis is saved, so the bank sees the trend, not just today's number
- **NEW — Bank Portal (Customer 360):** one screen per MSME for the credit officer — score, GST, bank, ITR, MCA and directors
- Integrity check: GST, ITR and bank credits are cross-checked; mismatches are flagged for officer review, not silently scored
- Built for Bharat: 7 Indian languages (Bhashini), AI assistant, printable & emailable Financial Report; score in seconds; API-ready for LOS / ULI / OCEN

Opportunities

How is it different from existing solutions?

- Bureau scores (CIBIL / Experian) need credit history — we score NTC / NTB MSMEs with none, from their own digital footprint
- Five data sources in one consent flow — Udyam, GST, ITR, bank (AA) and MCA — instead of a single-source view
- Score + score history + bank portal: a working system a credit officer can use today, not just a number
- Cross-source check (GST vs ITR vs bank credits) as a built-in integrity signal — no existing tool does this

How does it solve the problem?

- Ends the 'no documents, no credit' cycle — the MSME's existing filings and statements are the documents
- 2–3 weeks of manual appraisal becomes an automated assessment in seconds
- Every score carries dimension-level reasons — the officer sees why, not just what
- Neutral baseline for first-time borrowers — missing history is not treated as bad history

USP of the proposed solution

- Six-pillar score (Revenue, Cash Flow, Transactions, Compliance, Stability, Debt) — a far richer picture than any single metric
- Built on India's DPI stack — consent-based, no proprietary data, aligned with the DPDP Act 2023
- Product recommender maps the score band to CGTMSE / MUDRA / standard loans — more approvals, safely
- Score history gives IDBI post-disbursement monitoring — origination and monitoring from one platform

List of features offered by the solution

Data Aggregation — 5 sources, one consent

- Udyam registry · GST (GSTR-1 / 2A / 2B / 3B, HSN, credit notes, ITC) · ITR (ITR-4 / 5 / 6) · Account Aggregator bank statements (Finvu) · MCA company, charges & DIN-verified directors
- GST ↔ ITR ↔ bank-credit reconciliation with a cross-source anomaly flag for officer review

AI / ML Scoring Engine

- Six weighted pillars, 0–1000: Revenue Vitality 25% · Cash Flow Health 20% · Transaction Trust 15% · Compliance 15% · Business Stability 15% · Debt Serviceability 10%
- Rules engine + ML.NET LightGBM calibration (0.6 / 0.4), SSA cash-flow trend, PFI explanations, neutral baseline when a source is missing

Financial Health Card & Report

- Score gauge, 6-pillar radar, top strengths & risks, lending eligibility — working-capital limit (Nayak) + term-loan capacity, 14-ratio appraisal
- Product recommendation (CGTMSE / MUDRA / standard), Score History trend, printable Financial Report, email delivery

Bank Portal — Customer 360 (new for the Finals)

- Portfolio list with scores & bands, bank dashboard, per-MSME tabs: Score, GST, Bank (AA deep analysis), ITR, MCA, Directors

Integration, Security & Language

- REST scoring API (ULI DSP-compatible contract) for LOS / OCEN; on-demand re-analysis from the MSME's blob folder
- Email-OTP login, CSP security headers, source-IP fraud audit; 7 Indian languages via Bhashini; Groq AI assistant

What's new since Round 1 — built for the Finals

Score History & Trend

- Every analysis is saved (append-only), never overwritten
- Trend chart on the Health Card and in the bank portal
- The base for post-disbursement early warning

Bank Portal · Customer 360

- Portfolio list with scores & bands, bank dashboard
- Per-MSME tabs: Score, GST, Bank, ITR, MCA
- Director drill-down with DIN verification

MCA + DIN — 5th data source

- Company status, capital, charges & AGM compliance
- Directors verified against DIN with individual PAN
- Follows entity type — shown only for companies

ITR now inside the score

- Filing timeliness, e-verification, tax paid ratio
- 44AB audit and tax-demand signals
- GST ↔ ITR turnover reconciliation

Lending & Financial Report

- Working-capital limit (Nayak) + term-loan capacity
- 14-ratio appraisal, bank-statement deep analysis
- Printable Financial Report with email delivery

Demo-ready & Bharat-ready

- 5 seeded MSMEs covering 813 → 376
- IDBI palette, redesigned score gauge
- 7 languages (Bhashini), Groq AI assistant

Process flow diagram or Use-case diagram

1 · Onboard & log in

1. Enter Udyam number

Udyam registry lookup fetches the enterprise profile

2. Review & consent

Owner confirms details, email + mobile, gives consent and registers (mobile seeded in the Finvu SimBanks sandbox)

3. Email OTP → dashboard

OTP verified, session starts, routed by status

Existing MSME

/Auth/CustLogin → email → OTP → dashboard

2 · Collect data (one button each)

4a. GST

12 months × 7 return files — GSTR-1, 2A, 2B, 3B, HSN, credit notes

4b. Bank statements (AA)

Finvu consent → owner approves → FI fetch of the statement

4c. ITR

Filing, income, tax paid, audit

4d. MCA + DIN

Companies only — status, charges, directors

Stored per MSME

each source saved as JSON in the <UAN>/blob folder; fetched sources are locked

3 · Score

5. Analyze

Five extractors read every file; a missing source redistributes its weight (neutral baseline)

6. Rules 0.6 + LightGBM 0.4

Six dimensions · 0–1000 · band · PFI reasons · GST-ITR-bank anomaly · lending eligibility & products

7. Persist

result.json → Blob · score summary → SQL
· score history appended → SQL

Re-analyze

any time — re-runs 5–7 from the stored files; every run adds a history point

4 · Use the score

8. Financial Health Card

Gauge, 6-pillar radar, strengths & risks, products, Score History

9. Financial Report

Printable appraisal — print or email

10. Bank Portal

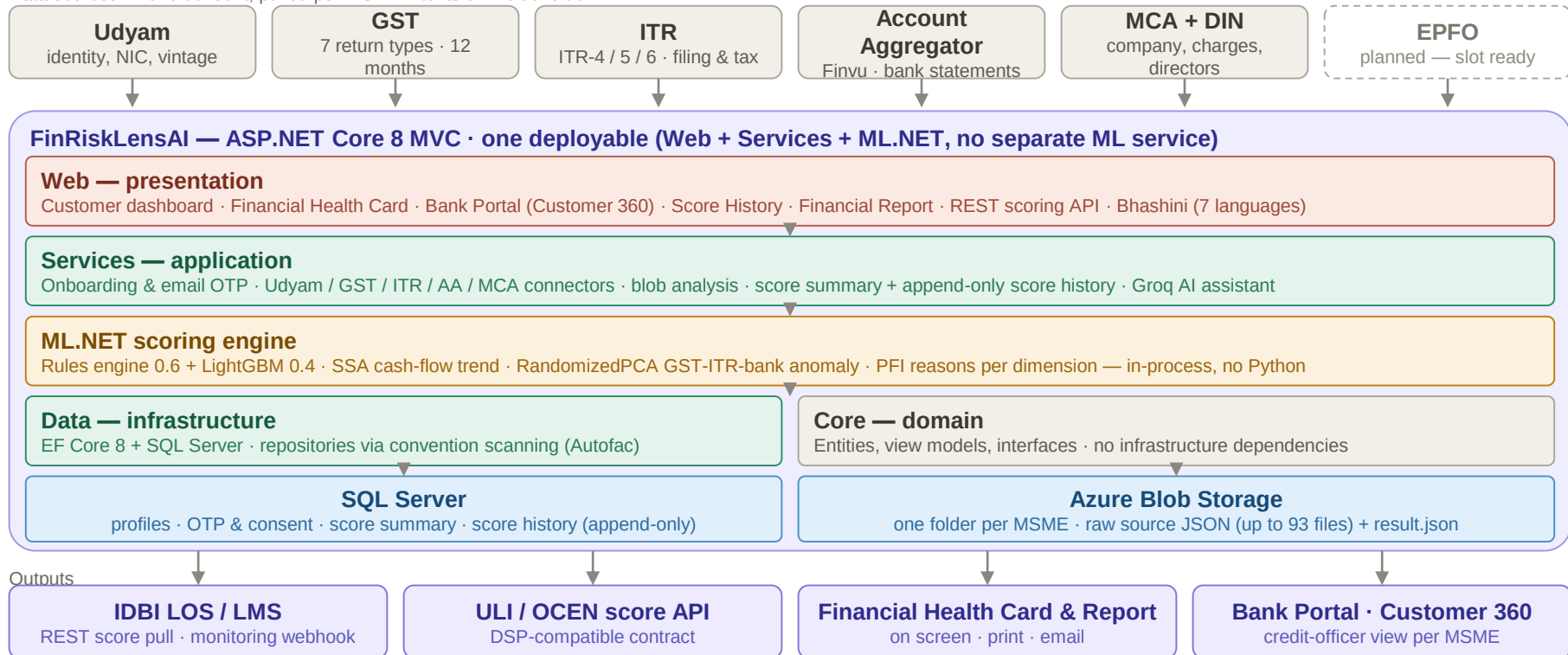
Bank login → customer list → Customer 360: Score, GST, AA, ITR, MCA, Directors

11. API

LOS / LMS pull · ULI / OCEN score API

Architecture diagram of the proposed solution

Data sources — one consent, pulled per MSME into its own blob folder



Technologies to be used in the solution

Backend — Orchestration & API

- ASP.NET Core 8 MVC (C#) — connectors for Udyam, GST, ITR, Account Aggregator (Finvu), MCA / DIN; REST scoring API
- ML.NET in-process — scoring, feature extraction and explanations in the same deployable; no separate ML service
- Autofac DI with convention scanning · Serilog structured logging · CSP security headers · DataProtection keys persisted

AI / ML Stack

- ML.NET LightGBM calibration — final score = 0.6 rules engine + 0.4 ML model, trained on 3,000 MSME profiles
- ML.NET SSA time-series — 12-month cash-flow trend · RandomizedPCA — GST vs ITR vs bank anomaly detection
- Permutation Feature Importance — per-dimension, audit-ready reasons · Groq LLM — AI assistant for questions on the score

Data & Storage

- SQL Server + EF Core 8 — MSME profiles, OTP & consent, score summary and append-only score history; SQL-backed sessions
- Azure Blob Storage — one folder per MSME: raw source JSON (up to 93 files) + result.json

Infrastructure & DevOps

- Single deployable web app (modern monolith) — IIS / Azure App Service / AWS; GitHub source control
- Planned: JWT-secured API + OpenAPI (Swagger) spec, Azure API Management as the ULI / OCEN gateway

Technologies to be used in the solution

Frontend

- ASP.NET Core MVC + Razor views, Bootstrap 5 — server-rendered, responsive; two themes including IDBI's official palette
- Chart.js — score gauge, 6-pillar radar, score-history trend, cash-flow charts · ApexCharts — GST analysis; charts follow the theme switch
- Bhashini website-translation plugin — English + Hindi, Marathi, Tamil, Gujarati, Konkani, Sanskrit

DPI & Data Integrations

- Account Aggregator (Finvu sandbox, SimBanks FIP) · GST System API · Udyam Registry · ITR · MCA 21 (company, charges, DIN)
- Unified Lending Interface (ULI) DSP API and OCEN 2.0 — API contract ready · EPFO REST API — planned

Security

- Email-OTP customer login and separate bank-admin login, SQL session cache, CSP headers, source-IP fraud / VPN / Tor audit with geolocation

Estimated implementation cost (optional)

₹ 22 L

One-time setup (build + security)

₹ 31 L / yr

Recurring (cloud + data + support)

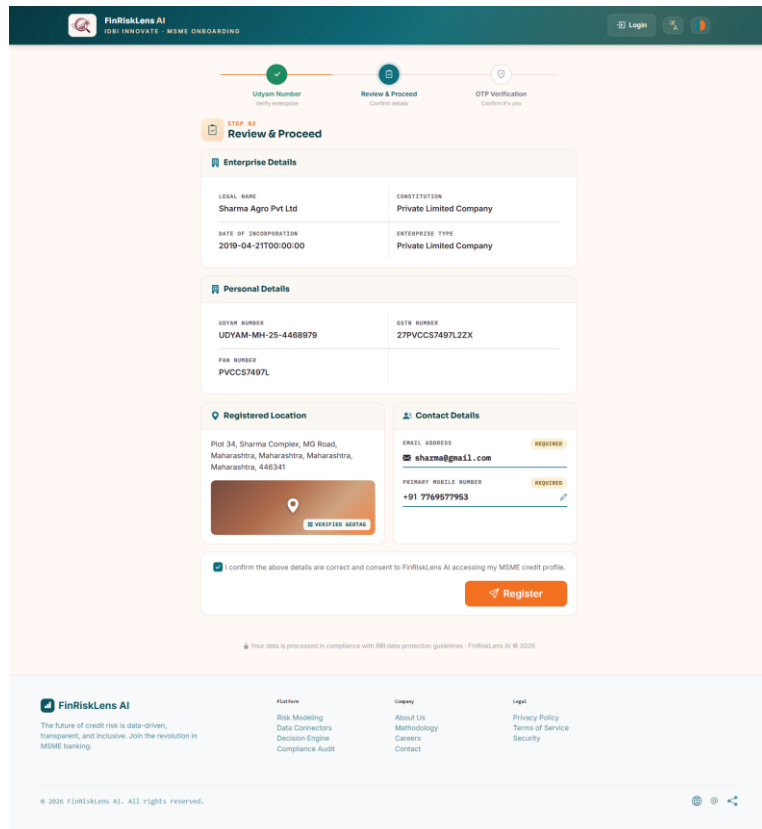
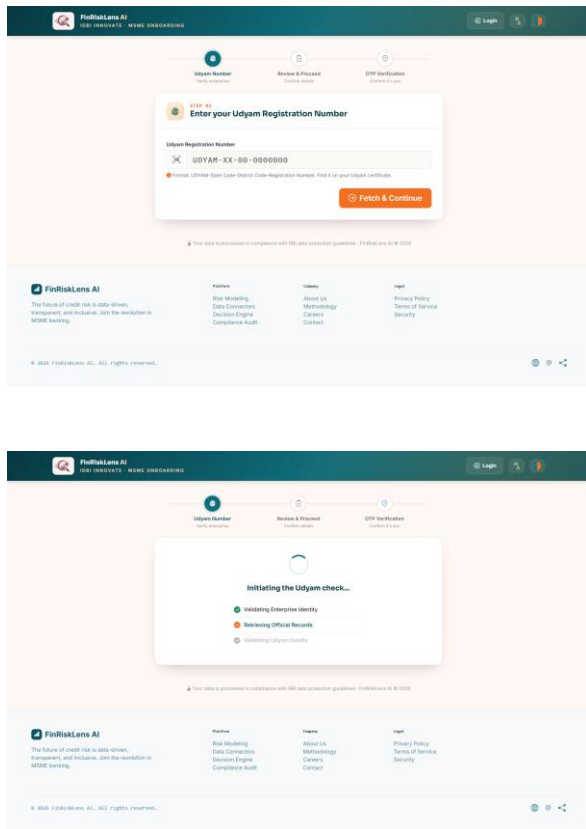
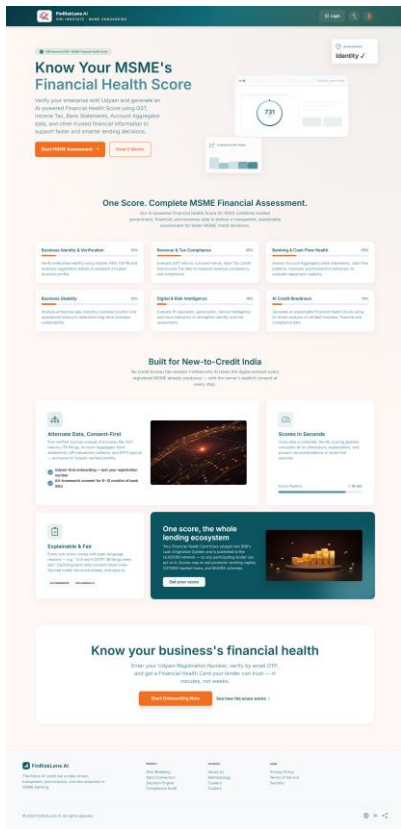
~ ₹ 1.15 Cr

3-year total cost of ownership

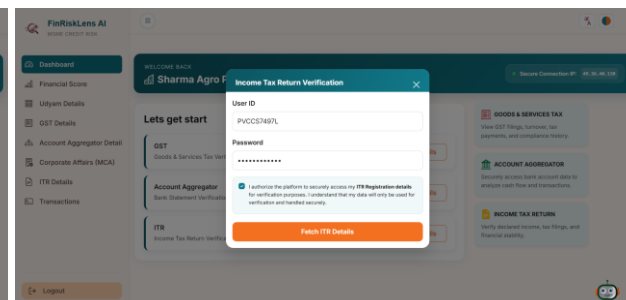
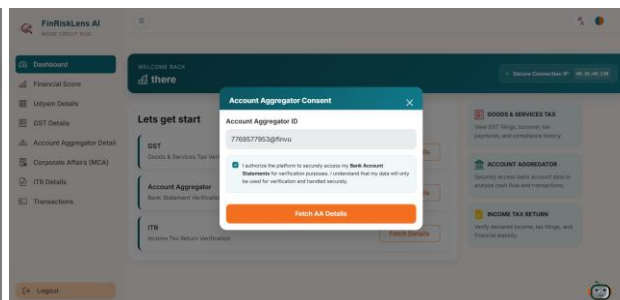
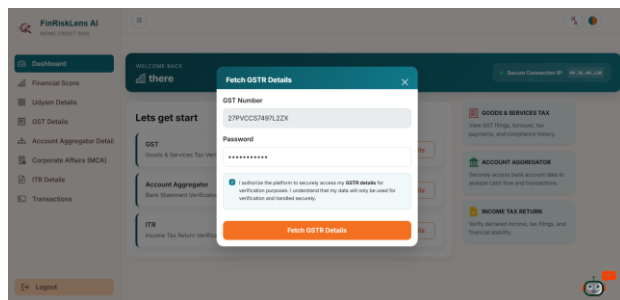
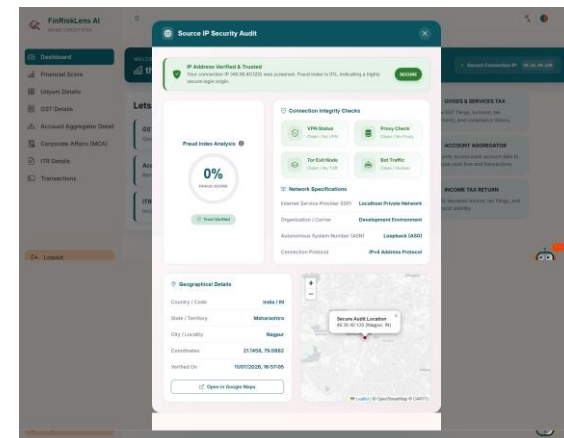
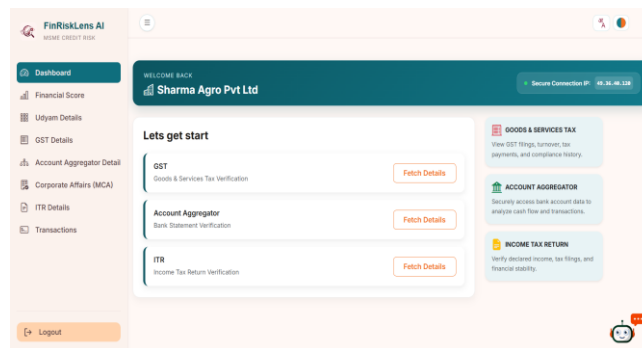
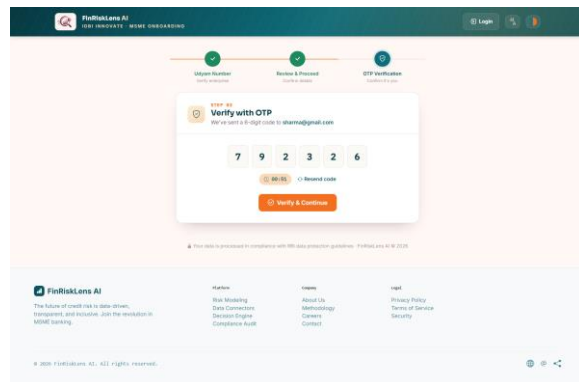
Cost breakdown - centralized cloud deployment serving all IDBI branches pan-India

Cost head	Type	Estimate (INR)
Development - 3-member team, ~6-month build	One-time	₹ 18 L
Security testing & RBI / DPDP compliance audit	One-time	₹ 4 L
Cloud & servers - Azure App Service, SQL, Blob, APIM	Annual	₹ 12 L
Data & API usage - AA (Finvu), GST, ITR	Annual	₹ 10 L
Maintenance, support & ML model retraining	Annual	₹ 9 L

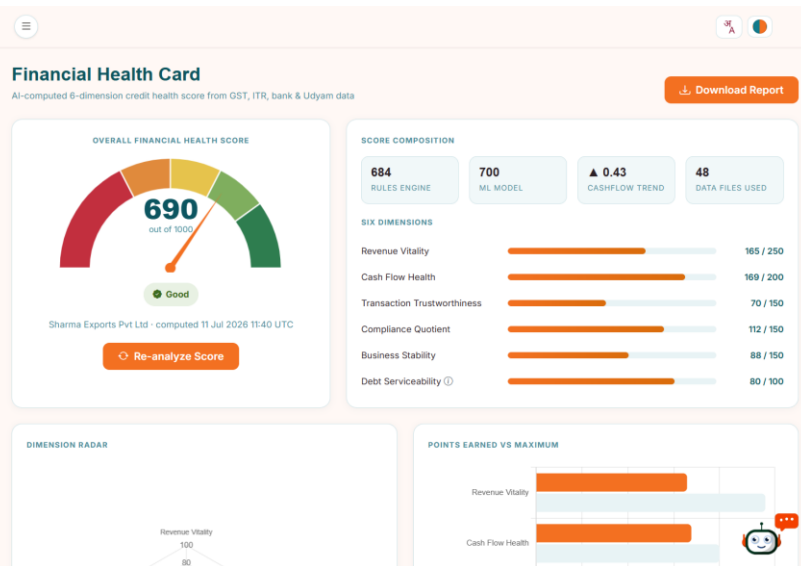
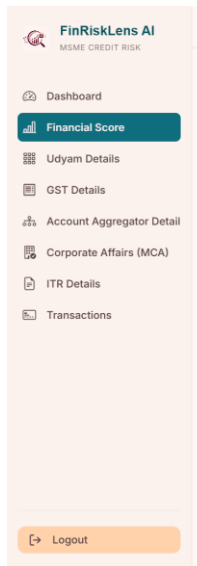
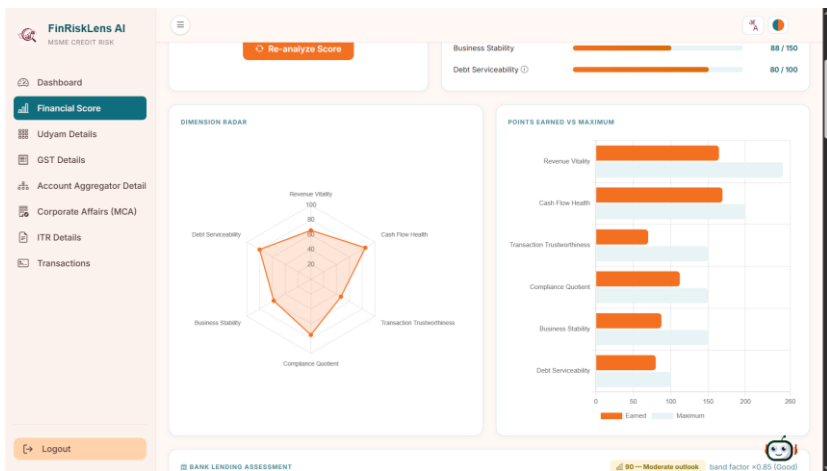
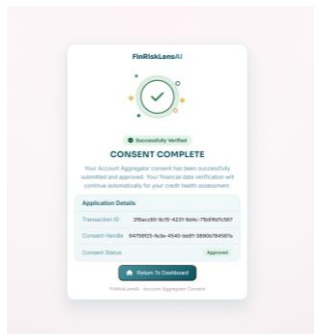
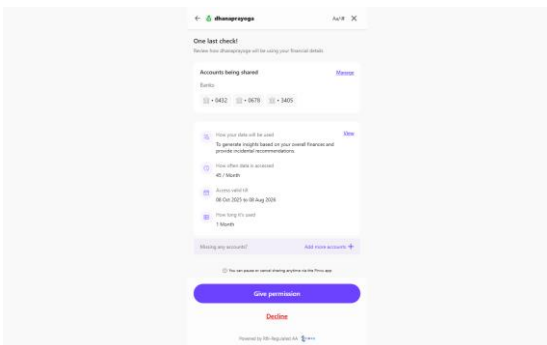
Snapshots of the prototype



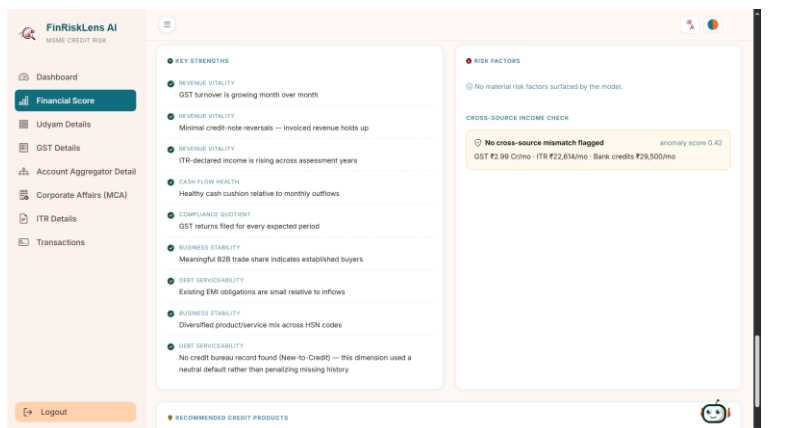
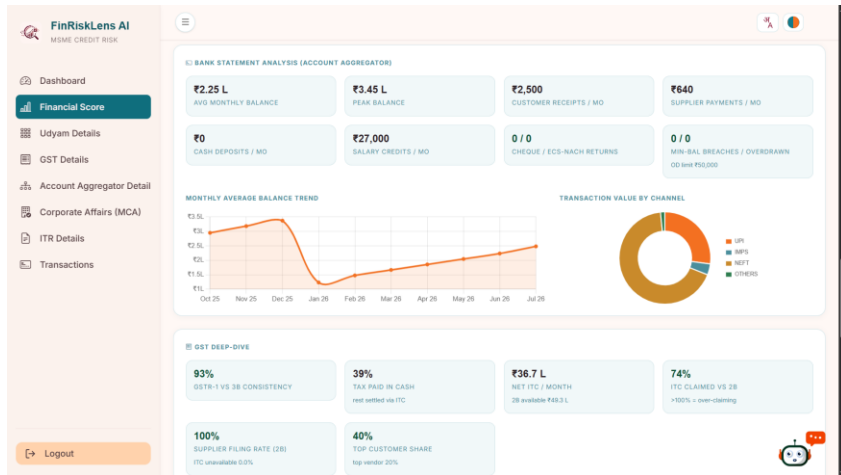
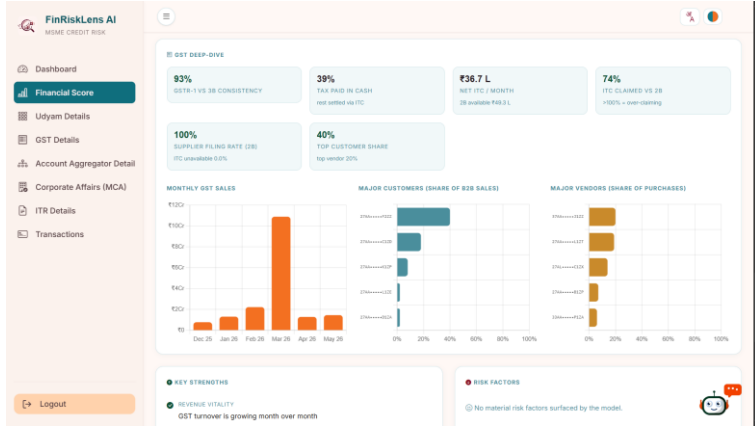
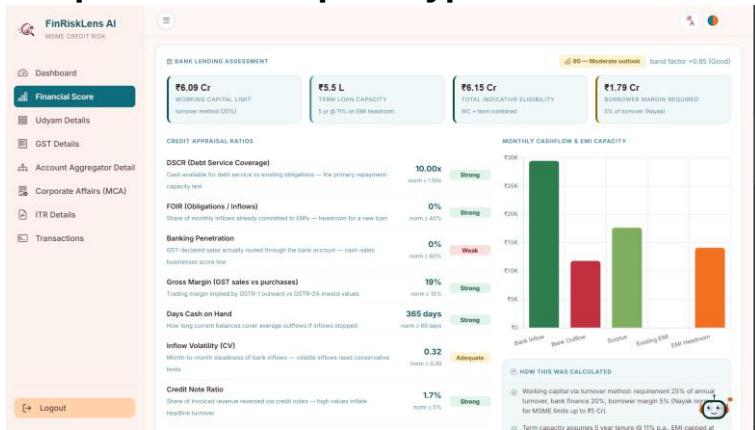
Snapshots of the prototype



Snapshots of the prototype



Snapshots of the prototype



Snapshots of the prototype

FinRiskLens AI

MSME CREDIT RISK

Dashboard

Financial Score

Udyam Details

GST Details

Account Aggregator Detail

Corporate Affairs (MCA)

ITR Details

Transactions

Logout

REVENUE VITALITY

ITR-declared income is rising across assessment years

GST ₹2.99 Cr/mo - ITR ₹22,644/mo - Bank credits ₹28,500/mo

CASH FLOW HEALTH

Healthy cash cushion relative to monthly outflows

COMPLIANCE QUOTIENT

GST returns filed for every expected period

BUSINESS STABILITY

Meaningful B2B trade share indicates established buyers

NET SERVICEABILITY

Existing EM obligations are small relative to inflows

BUSINESS STABILITY

Diversified product/service mix across HSN codes

NET SERVICEABILITY

No credit bureau record found (New-to-Credit) — this dimension used a neutral default rather than penalizing missing history

RECOMMENDED CREDIT PRODUCTS

Standard Working Capital Loan

₹5.38 Cr — ₹10.75 Cr

Good financial health score supports a standard working capital facility

Business Term Loan

₹7.17 Cr — ₹14.34 Cr

Consistent revenue and compliance history support term lending

Model fit-scoring v1.0 - Indicative amounts anchored to observed turnover — not a sanctioned limit.

FinRiskLens AI

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GST RETURNS VERIFIED

AI-Powered GST Intelligence Analysis

Advanced GSTR-2B & GSTR-3B Business Intelligence Dashboard • Sharma Agri Pvt Ltd

GSTRIN

PAN

UDYAM REGISTRATION

BUSINESS AGE

INDUSTRY

ENTERPRISE TYPE

GST HEALTH SCORE

94

Excellent GST Compliance

GST HEALTH SCORE

94

94/100

GST COMPLIANCE

98%

Consistent

BUSINESS GROWTH

87%

Consistent

RISK SCORE

Low

Stable

COMPLIANCE RATING

A+

Excellent

GST HEALTH

Healthy

Passed Analysis

BUSINESS STABILITY

94%

84% Stability

CASH FLOW

Healthy

Positive Operating CF

AI Business Assessment Engine

AI ASSESSMENT RATIONALE

Business has maintained excellent GST compliance (98%) for the last 24 months.

Revenue growth is highly consistent (+18% outward supply YoY).

Vendor dependency is low (highest customer)

STABLE PERFORMANCE

COMPLIANCE LEVEL

GST Compliance & Filing History

REGISTRATION ACTIVE

9-MONTH FILING TIMELINE (GSTR-1 & 3B)

Jan

Feb

Mar

Apr

May

Jun

Jul

Aug

Sept

Oct

Nov

Dec

RETURNS

LATE

COMPLIANCE

REGISTRY

FinRiskLens AI

MSME CREDIT RISK

Dashboard

Financial Score

Udyam Details

GST Details

Account Aggregator Detail

Corporate Affairs (MCA)

ITR Details

Transactions

Logout

Corporate Intelligence Report

85

100

Active

2

8

₹3.72 Cr

LOW

Overview

Directors

Charges

Risk Analysis

CAPITAL STRUCTURE

Authorized

₹19 L

Paid Up

₹44 L

Members

16

COMPLIANCE

AGM

16 Mar 2022

Balance Sheet

16 Mar 2022

CIRP

No

ROC

RoC-Maharashtra

Corporate Profile

Company Name

Sharma Exports Pvt Ltd

CIN

U85074MH2021PTC524370

Type

Private Company

Category

Company limited by Shares

Status

Active

Listed

No

Incorporated

16 Mar 2021

AI Observations

Company is Active

Private Limited Company

Registered under RoC-Maharashtra

No CIRP proceedings found

Company has active secured borrowings (8 open charges)

Proceed with GST & Banking assessment

Incorporation Journey

Company Incorporated

16 Mar 2021: RoC-Maharashtra

Directors Appointed

3 directors added

First Charge Created

2022: Secured borrowing initiated

Charges Modified

2022 - 2026: Multiple modifications

Current Status: Active

Today: Company operational

Quick Risk Indicators

COMPANY

Active

LISTED

No

CIRP

No

SUSPENDED

No

OPEN CHARGES

8

DIRECTORS

Stable

Corporate credibility is Low (85/100). Active company with 2 verified directors and secured financial charges (₹16.85 Cr open). Suitable for further GST and Banking assessment.

Snapshots of the prototype

FinRiskLens AI

MSME CREDIT RISK

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UDYAM REGISTRATION ACTIVE

AI-Powered Udyam Registration Analysis

Sharma Agro Pvt Ltd - Registration No: UDYAM-MH-25-4468979

Download Report

Refresh Data

BUSINESS IDENTITY SCORE

73 / 100

Medium Risk Growing Business

Business Identity

Business Name: Sharma Agro Pvt Ltd

Enterprise Type: Micro

Business Age: 7 Years

Registration Date: 06 May 2019

Organization Type: Private Limited Company

Major Activity: Manufacturing

Identified Strengths

Registered MSME

Operating since 2019 (+7 Years)

Stable Micro Enterprise Classification

GST Registered

PAN Verified

AI Observations

Established MSME operating for over 7 years with a consistently maintained Micro Enterprise classification across all recorded years. Multiple registered plant locations (1), verified GST and PAN records, and diversified NIC service activities point to a mature, credible business profile. Overall business identity confidence is assessed as Medium Risk, supporting a favorable credit assessment posture.

Enterprise Classification Timeline

2019 MICRO

LOCATION SUMMARY

State: Maharashtra

District: Maharashtra

City: Maharashtra

PIN: 267084

Total Locations: 1

COMPLIANCE SUMMARY

GST: 27XXXXXXX022X

PAN: XXXXX687L

Registration: Active

Registered Plant Locations (1)

UNIT NAME: Sharma Agro Pvt Ltd

CITY: Maharashtra

DISTRICT: Maharashtra

STATE: Maharashtra

NIC Activities (1)

NIC CODE: 81967

ACTIVITY: Manufacturing

TYPE: Manufacturing

FinRiskLens AI

MSME CREDIT RISK

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WELCOME BACK

Sharma Agro Pvt Ltd

Secure Connection IP: 49.36.48.138

Lets get start

GST Goods & Services Tax Verification

Fetch Details

Account Aggregator Bank Statement Verification

Fetch Details

ITR Income Tax Return Verification

Fetch Details

FinRiskLens AI Assistant

Online

Hi! I'm your FinRiskLens AI assistant. I can explain anything on the FinRiskLens AI dashboard — scores, statuses, what the numbers mean. What would you like to know?

Ask about this page...

FinRiskLens AI

IDBI INNOVATE - MSME ONBOARDING

Welcome Back!

Enter your Email ID

Enter your registered email

We'll send a 6-digit OTP to your registered email.

Generate OTP

English

Hindi (हिन्दी)

Marathi (मराठी)

Tamil (தமிழ்)

Gujarati (ગુજરાતી)

Goan Konkani (गोंय का कोंकणी)

Sanskrit (संस्कृत)

Powered By

FinRiskLens AI

The future of credit risk is data-driven, transparent, and inclusive. Join the revolution in MSME banking.

Platform

Risk Modeling

Data Connectors

Decision Engine

Compliance Audit

Company

About Us

Methodology

Careers

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Legal

Privacy Policy

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Security

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Prototype Performance report/Benchmarking

715/1000

Live score on a real MSME — 12 months of GST + bank data (band: Good)

< 10 s

End-to-end scoring (sub-second when warm)

5

Data sources fused: Udyam · GST · ITR · AA · MCA (+ EPFO ready)

3000

MSME profiles the LightGBM model trains on

Benchmark vs. traditional credit assessment

Metric	Traditional / bureau	FinRiskLensAI
Turnaround time	2-3 weeks (manual)	Seconds (automated)
NTC / NTB borrowers	Rejected - no history	Scored on alternate data
Data sources	1 - bureau file	5 - Udyam, GST, ITR, AA, MCA (+ EPFO ready)
Explainability	Black-box score	PFI reason per dimension
Fraud / integrity check	None	GST-ITR-bank anomaly + MCA / DIN checks
Monitoring	One-time assessment	Score history + trend on every analysis

Additional Details/Future Development (if any)

Next Data Integrations

- EPFO — employee count & PF-remittance regularity (the feature slot is already built)
- GST e-invoice / e-way bill signals and CERSAI collateral lookup
- ONDC / GeM / UPI merchant-settlement data for micro-enterprises

Monitoring, Fraud & Compliance

- Early Warning System on score history — band drop, GST filing lapse, new MCA charge, bounce spike
- Sanctions / PEP screening, RBI fraud registry & wilful-defaulter match
- DPDP consent ledger across all sources with customer-side revoke & auto-purge

Lending Ecosystem & Integration

- JWT-secured API with OpenAPI (Swagger) spec — LOS score pull, LMS monitoring webhook
- ULI DSP + OCEN 2.0 certification; CIBIL / CRIF bureau blend for NTB borrowers
- GST-invoice financing and co-lending workflows with NBFC partners

Decision & Governance

- Credit-policy console — Approve / Refer / Decline with editable product-wise cut-offs & knock-outs
- PDF Credit Appraisal Memo with data-provenance appendix (source, time, consent ID)
- Model card & governance page (RBI model-risk); portfolio analytics for IDBI teams

Provide links to your:

- GitHub Public Repository : <https://github.com/abhinavmukwane/FinRiskLensAI>
- Demo Video Link (3 Minutes): <https://video.finrisklensai.com> OR [Google Drive Link](#)
- Final Product Link: <https://finrisklensai.com>

Demo accounts — customer login at /Auth/CustLogin (enter email → OTP is shown on screen) · bank portal at /Auth/BankLogin, user bankadmin

UDYAM-MH-20-0091447 · 813 Excellent · Shreeji Precision Components · shreeji.precision@finrisklens.demo

UDYAM-MH-25-0042817 · 806 Excellent · Sankalp Autotech Pvt Ltd (full MCA + DIN) · sankalp.autotech@finrisklens.demo

UDYAM-MH-20-0067394 · 715 Good · M/s UCN Fibrenet Pvt Ltd (real filings) · mkhangar01@gmail.com

UDYAM-GJ-01-0044219 · 627 Fair · Ratnadeep Textile Traders · ratnadeep.textiles@finrisklens.demo

UDYAM-UP-28-0007733 · 376 At Risk · Navdeep Auto Spares · navdeep.autospares@finrisklens.demo

The five together cover the whole score range — approve, judge, and decline — with every data tab populated.



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Thank you!